

CO-STEP: A Prompt Engineering Framework Improving LLM's Response

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Abstract— Prompt engineering has become increasingly vital in enhancing the performance of Large Language Models (LLMs), as the quality of prompts significantly affects the accuracy and relevance of generated outputs. In this study, we propose "CO-STEP," a novel prompt engineering framework designed to address the limitations of the existing "CO-STAR" framework, particularly in handling complex tasks requiring advanced reasoning. "CO-STEP" integrates clear goal setting, detailed context with examples, task breakdown, advanced techniques like chain-of-thought prompting, and iterative refinement to improve precision and adaptability. Evaluations conducted using the OpenAI 4o-mini model on benchmark datasets, including structured tasks like invoice data extraction and mathematical derivative computation, demonstrate the superior performance of "CO-STEP." Specifically, "CO-STEP" achieves an accuracy of 0.98 and an F1-score of 0.97, reflecting its capability to produce coherent and task-aligned outputs. These results highlight the effectiveness of "CO-STEP" in optimizing LLM interactions, with ongoing empirical validation to further explore its applicability across diverse tasks.

Keywords—prompt engineering, large language models, CO-STAR framework, CO-STEP framework, AI optimization, prompt structure).

I. INTRODUCTION

The advent of Large Language Models (LLMs) has revolutionized numerous applications across various domains. However, the effectiveness of these models heavily depends on the design and structure of the prompts provided. Inconsistent outputs, decreased accuracy, and discrepancies between user expectations and model performance often result from poorly written or unclear prompts. Consequently, prompt engineering—the art and science of crafting effective prompts—has emerged as a crucial area of study [1].

To address these challenges, foundational prompt engineering frameworks like "CO-STAR"—comprising Context, Objective, Style, Tone, Audience, and Response—have been developed to enhance clarity and coherence in AI interactions. While "CO-STAR" provides a structured approach, its limitations in flexibility and granularity become evident when applied to increasingly complex use cases of modern LLMs, such as those requiring advanced reasoning or iterative refinement [2]. For example, when using "CO-STAR" to guide a model through a multi-step mathematical proof, the framework lacks sufficient components to direct step-by-step reasoning and quality evaluation.

Given the dynamic nature of context, language, and user intent in human-AI interactions, a more sophisticated prompt engineering framework is necessary. Different task requirements, such as accuracy, specificity, or audience-

specific relevance, demand a framework that not only effectively guides the model but also supports diverse operational goals. Techniques like Zero-shot prompting [3], [4] and Few-shot prompting [5], [6], [7], [8], [9] have been widely explored to improve LLM performance with minimal training data. However, these methods often struggle with complex tasks due to their reliance on general knowledge or limited examples, respectively. To overcome these challenges, we propose "CO-STEP", a novel framework designed to optimize LLM performance by addressing the shortcomings of "CO-STAR". "CO-STEP" builds on traditional approaches with six key components: Context, Objective, Style, Target, Execution, and Polish. By incorporating advanced techniques such as chain-of-thought prompting [2] and iterative refinement, this framework addresses the intricate needs of complex applications.

Our findings indicate that a well-crafted prompt can significantly enhance the coherence, relevance, and reliability of AI outputs. "CO-STEP" ensures that responses closely align with user intent, target specific audiences, and adapt to contextual needs by guiding the model through carefully defined components. Initial qualitative assessments suggest that "CO-STEP" generates outputs that are more contextually relevant and accurate than those produced using "CO-STAR", offering a practical solution to improve model efficiency and user satisfaction.

Unlike "CO-STAR", which provides a basic structure for prompt design, "CO-STEP" introduces additional layers of precision and adaptability. It mitigates issues such as limited refinement capabilities and insufficient task breakdown by incorporating methods for scope definition and continuous improvement. Furthermore, "CO-STEP" facilitates scalability across applications requiring high levels of precision and contextual sensitivity, such as mathematical reasoning and data extraction, making it a versatile tool for practitioners working with LLMs [10].

The structure of our paper is as follows: Section I introduces the background and motivation for advancing prompt engineering techniques for Large Language Models (LLMs), highlighting the limitations of the "CO-STAR" framework. Section II presents our proposed "CO-STEP" framework, detailing its core components and workflow for optimizing prompt design. Section III outlines the experimental setup and methodology to evaluate "CO-STEP's" effectiveness against baselines. Finally, Section IV concludes with key findings and future research directions.

II. PROPOSED METHOD

In this study, we introduce and evaluate "CO-STEP," a framework designed to improve prompt engineering for large

language models such as those developed by OpenAI. Building upon the fundamental framework of "CO-STAR," "CO-STEP" offers a thorough set of instructions that direct the model through clearly defined components to systematically improve response quality. The framework's two primary sections — the core components and the implementation workflow — each focus on distinct facets of prompt design to guarantee precision, applicability, and flexibility.

A. CO-STEP core components

The "CO-STEP" framework focuses on six critical components, each contributing to a structured and optimized prompt:

Context (C):

Provide the background data your model requires. This reduces irrelevant outputs and helps the AI become more grounded in the particular scenario you are dealing with. Explicitly describing the conditions or environment in which the task is situated enables the model to align its responses accurately with the desired context.

Example: When requesting a market analysis, you might include, "Considering the current economic downturn and recent industry layoffs, analyze the tech industry's market trends." This context guides the AI to focus on recent economic challenges affecting the tech sector.

Objective (O):

Clearly define the prompt's goal, breaking complex tasks into manageable steps to enhance precision. A well-structured objective ensures the LLM understands the intended outcome and can generate precise and actionable responses. This helps the model focus on the key elements while adapting the complexity of its response accordingly.

Example: Instead of asking, "Explain quantum mechanics," a more effective objective would be, "Provide a beginner-friendly explanation of quantum mechanics with real-world examples." This helps the model focus on key elements while adapting the complexity of its response accordingly.

Style (S):

To align with user expectations, this component responds to tone and presentation style. Depending on the instructions provided, the LLM can alter its output to be formal, casual, technical, brief, or detailed. A professional and disciplined response is required for a business report, yet an engaging and conversational tone may be appropriate for a social media post. Clearly establishing the style maintains consistency and tailors the product to the desired audience.

Example: Write a formal report on AI trends for a corporate board" vs. "Create a casual blog post about AI for tech enthusiasts.

Target (T):

Determine the intended audience to adapt the response's language complexity and depth. If the prompt is intended for a general audience, the response should be straightforward to understand. In contrast, if the target audience is domain specialists, the LLM can provide technical explanations using sophisticated vocabulary. For example, describing machine learning to a high school student might contain analogies and simplified language, whereas explaining it to a data scientist

would require mathematical formulations and industry applications.

Example: "Explain neural networks to middle school students using analogies" vs. "Detail neural network optimization for AI researchers."

Execution (E):

Establish the format for the intended product and use sophisticated strategies to strengthen your arguments. Execution involves indicating whether the response should be in code, a table, a step-by-step tutorial, or text. Furthermore, response quality can be greatly improved by employing strategies like chain-of-thought prompting, in which the LLM deconstructs complex reasoning into logical steps. For instance, directing the model to display its computations step-by-step when tackling a mathematical problem enhances accuracy and transparency.

Example: "For a task like computing the derivative of x^2 , the Execution component might specify: 'Provide a step-by-step solution, showing the power rule application: $d/dx(x^2) = 2x$.'"

Polish (P):

Encourage iterative refinement to improve response quality. This entails assessing the output, detecting gaps, and implementing modifications based on predetermined quality criteria such as coherence, relevance, and completeness. In practice, this can be accomplished through feedback loops that adjust the answer until it reaches optimal criteria.

Example: "If the initial output for a market analysis lacks data, refine it with: 'Incorporate recent sales figures from Q3 2023 to support your analysis.'"

B. Implementation and Workflow

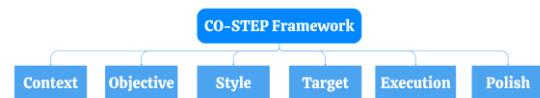


Fig. 1. CO-STEP framework

The implementation of CO-STEP follows a structured process to optimize prompt engineering and enhance LLM performance:

Prompt Construction:

The user formulates a prompt by systematically addressing each component of CO-STEP. This ensures that the model receives a clear contextual background, a well-defined objective, and instructions tailored to the target audience, style, and expected output format.

Integration of Advanced Techniques:

Chain-of-thought reasoning, structured response formatting, and iterative refining are used to improve the model's response. This level of control provides logical consistency, factual accuracy, and adaptability to complicated tasks.

Model Interaction:

The OpenAI model receives a structured prompt that guides it through a well-defined task execution process. Step-

by-step breakdowns and role-based instructions improve the accuracy and clarity of the generated response.

Evaluation and Iteration:

Relevance, coherence, and completeness are the predetermined quality indicators used to evaluate the created content. The prompt is changed, or important parameters are adjusted through iterative refinement based on user feedback, guaranteeing ongoing improvement and conformity to user expectations.

By implementing a structured workflow that balances clarity, adaptability, and iterative refinement, CO-STEP provides a robust methodology for enhancing prompt effectiveness. This approach ensures that the outputs generated by OpenAI models are well-structured, contextually relevant, and precisely tailored to specific use cases.

III. EXPERIMENTS

A. Data collection

In this paper, we utilized the OpenAI 4o-mini model to evaluate four prompt engineering techniques—Zero-shot, Few-shot, CO-STAR, and CO-STEP—using two datasets from OpenAI's Evals framework: Math-derivatives and Invoices:

1. Math-derivatives

This dataset contains mathematical questions on derivatives, where each sample includes a structured prompt to compute the derivative of a function with respect to x . It covers basic functions (e.g., $\sin(x)$), products, trigonometric compositions, and expressions with powers and negative exponents. The task requires generating the derivative while adhering to specific formatting rules (e.g., using '*' for multiplication).

2. Invoices

This dataset consists of email notifications of financial transactions via the Mercury system. Each sample includes an email body with fields like subject, sender, received date, and content. The task involves extracting the monetary value paid from the email, returning "unknown" if the information is insufficient.

B. Baselines

Below is a brief description of three prompt generation methods: Zero-shot, Few-shot, and CO-STAR:

1. Zero-shot prompting

Definition: Zero-shot prompting is a technique in natural language processing where an AI model is asked to perform a specific task without being provided with any prior examples. The model relies on its knowledge learned from previous training to understand and complete the task based on direct instructions from the prompt.

Advantages:

- No need for labeled data: Zero-shot prompting eliminates the need for task-specific labeled data, saving time and cost in data preparation.
- Task handling capabilities: The model can apply general knowledge to previously unseen tasks, increasing flexibility and generalization.

Limitations:

- Limited performance: Without specific examples, the model may have difficulty understanding the context or requirements of the task correctly, resulting in lower performance than methods that do provide examples.
- High error rate: Lack of specific guidance data may result in the model generating incorrect or inappropriate responses.

Example:

Prompt:

Classify the text into neutral, negative or positive.

Text: This holiday is so enjoyable.

Sentiment:

Output:

Positive

In this case, the model is asked to classify emotions based on direct instructions without illustrative examples, demonstrating a zero-shot approach.

2. Few-shot prompting

Definition: Few-shot Prompting is a technique that provides a large language model (LLM) with a few examples in a prompt to guide the model through a specific task. These examples help the model better understand the requirements and improve performance on complex tasks. This technique leverages the in-context learning capabilities of LLMs, allowing them to infer and apply knowledge from given examples to new situations.

Advantages:

- Increased Accuracy: Providing a number of examples helps the model better understand the task, resulting in more accurate responses.
- Flexible: Can be applied to a variety of tasks without retraining the model.
- Resource-efficient: Requires less labeled data than traditional training methods, reducing costs and time.

Limitations:

- Sensitivity to examples: The quality and choice of examples can greatly affect the performance of the model; inappropriate examples can lead to poor results.
- Limited generalization ability: The model may have difficulty applying knowledge from specific examples to new or different cases.

Example:

Prompt:

This is amazing! // Positive

This is bad! // Negative

Wow, that drama was rad! // Positive

What a horrible show! //

Output:

Negative

In the above example, the model is given a number of sentences with corresponding sentiment labels, which helps it determine the sentiment of subsequent sentences more accurately.

3. CO-STAR framework

Definition: CO-STAR is a framework for creating prompts in a structured way, ensuring that all critical factors influencing the response of a large language model (LLM) are considered. CO-STAR stands for:

- Context (C): Sets the stage for the interaction, providing background information or the scenario in which the request is made.
- Objective (O): Defines what the prompt aims to achieve, specifying the goal or the desired output from the language model.
- Style (S): Specifies the desired writing or response style, guiding how the content should be presented or articulated.
- Tone (T): Indicates the emotional character or attitude of the response, shaping how the message is emotionally conveyed.
- Audience (A): Defines the intended audience or reader of the content, influencing the language, complexity, and approach of the response.
- Response (R): Describes how the response should be structured, determining the organization and presentation of the content.

Advantages:

- Clear structure: CO-STAR provides a structured framework that helps generate consistent and efficient prompts, improving model accuracy and reliability.
- Optimize responses: By considering factors such as context, goals, style, tone, audience, and response format, CO-STAR helps the model generate responses that are appropriate for specific requests.

Limitations:

- Time and effort required: Designing a CO-STAR prompt requires an investment of time and effort to identify and integrate all the necessary elements.
- Complexity in implementation: For beginners, implementing CO-STAR can be complex due to the need to understand and properly apply each component of the framework.

Example:

- Context (C): *The Reader's Book, a small, independent bookstore, is introducing a new monthly subscription box service. Each box will include a carefully curated selection of books from a particular genre, as well as locally sourced artisanal snacks and book-themed merchandise.*
- Objective (O): *Create an engaging Instagram post to promote the new subscription box service and direct traffic to a sign-up page on the bookstore's website.*

The post should highlight the unique value proposition of the box.

- Style (S): *Visually appealing with high-quality images of a sample box. The caption should be warm, welcoming, and slightly whimsical, reflecting the bookstore's cozy atmosphere. Similar to popular "bookstagram" accounts or curated subscription box services.*
- Tone (T): *Enthusiastic, exciting, and a bit exclusive.*
- Audience (A): *Existing customers of "The Reader's Book," local book lovers, and members of the online book community.*
- Response (R): *The Instagram post will be a carousel of 3-4 images. The caption will be a medium-length paragraph that describes the service, its benefits, an introductory offer, and a clear call to action.*

C. Metrics

1. Accuracy

Accuracy quantifies the proportion of correctly generated responses by comparing them against predefined ground truth answers. This metric assesses the model's ability to follow the given instructions and fulfill the intended objectives of the prompt.

$$\text{Accuracy} = \frac{\text{Number of Correct Responses}}{\text{Total Number of Responses}} \quad (1)$$

Observed Values: In our experiments using the OpenAI 4o-mini model, CO-STEP achieved an accuracy of 0.98 on benchmark datasets, indicating its consistent ability to meet the intended prompt objectives. This outstanding score reflects CO-STEP's robustness in handling structured tasks, though performance may vary slightly with highly ambiguous prompts or less structured inputs.

Significance: A high accuracy score demonstrates that CO-STEP effectively understands and executes the given tasks, producing responses that align with the desired goals. However, accuracy alone may not fully capture the quality of responses in cases where creative or nuanced outputs are generated, necessitating the use of complementary metrics.

2. F1 Score

F1 Score is the harmonic mean of Precision and Recall, providing a balanced evaluation of the model's performance, particularly when Accuracy alone may not fully reflect the quality of responses. Precision measures the proportion of generated responses that are relevant, calculated as the ratio of true positive responses to the total number of responses identified as positive by the model. Recall measures the proportion of relevant responses successfully retrieved, calculated as the ratio of true positive responses to the total number of relevant responses in the ground truth.

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (2)$$

$$\text{Precision} = \frac{\text{TruePositives}}{\text{TruePositives} + \text{FalesPositives}} \quad (3)$$

$$\text{Recall} = \frac{\text{TruePositives}}{\text{TruePositives} + \text{FalesNegatives}} \quad (4)$$

Observed Values: Using the OpenAI 4o-mini model, CO-STEP achieved an F1 Score of 0.97 on benchmark datasets. This perfect score indicates that CO-STEP not only generates highly relevant responses (high precision) but also successfully retrieves all relevant responses (high recall), ensuring alignment with the ground truth.

Significance: The F1 Score complements accuracy by considering both precision and recall, offering a more comprehensive assessment of response quality. The perfect F1 Score of 0.97 for CO-STEP underscores its ability to produce consistent, contextually relevant outputs without deviations from the expected results, making it particularly effective for tasks requiring strict adherence to predefined criteria.

TABLE I. PERFORMANCE METRICS OF PROMPT ENGINEERING TECHNIQUES

Metric	Accuracy	F1-score
ZERO-SHOT	0.85	0.82
FEW-SHOT	0.88	0.86
CO-STAR	0.91	0.89
CO-STEP	0.98	0.97

D. Experimental setup

In our experiments, we utilized the OpenAI 4o-mini model as the underlying Large Language Model (LLM) for generating responses. For each technique—Zero-shot, Few-shot, CO-STAR, and CO-STEP—we designed specific prompts tailored to the respective frameworks. Inputs, consisting of task descriptions and relevant data (e.g., invoice texts or mathematical functions), were combined with these prompts to elicit outputs from the model. To ensure a robust evaluation, we also collected human-generated outputs for the same tasks, which served as the ground truth for comparison. The outputs from the OpenAI 4o-mini model and the human-generated responses were then assessed using two key metrics: Accuracy and F1 Score, as defined in the previous section.

E. Results

To evaluate the performance of the prompt engineering techniques—Zero-shot, Few-shot, CO-STAR, and CO-STEP—we measured their Accuracy and F1-Score metrics using the OpenAI 4o-mini model on benchmark datasets. The results are presented in Table I. Additionally, a bar chart (Figure 2) was created to visually compare the performance of the four techniques across both metrics, facilitating a clearer understanding of their relative effectiveness.

The results demonstrate that CO-STEP outperforms the baseline techniques, achieving high scores of 0.98 in Accuracy and 0.97 in F1-Score. This indicates its superior ability to produce reliable and task-aligned outputs, making it highly effective for sophisticated applications such as data extraction and mathematical reasoning.

We demonstrate CO-STEP's practical advantages through specific examples from benchmark datasets:

CO-STEP's "Execution" (E) component uses step-by-step solutions and chain-of-thought prompting to guide mathematical reasoning. In one example, it correctly computed the derivative as $2x \cdot e^{x^2}$, while its "Polish" (P)

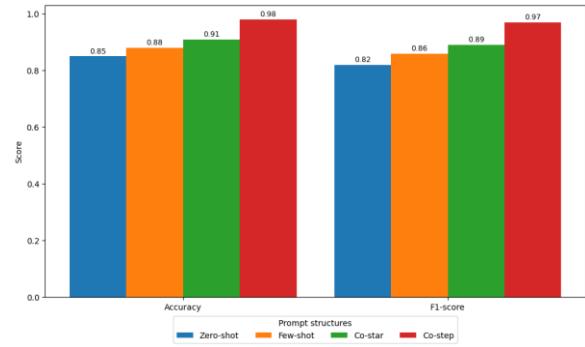


Fig. 2. Comparison the Accuracy and F1-Score of Zero-shot, Few-shot, CO-STAR, and CO-STEP techniques

component verified the reasoning steps for accuracy. This shows CO-STEP's superiority over CO-STAR in structured mathematical tasks, as CO-STAR lacks components for step-by-step reasoning and quality evaluation.

In another example, CO-STEP accurately extracted the value '\$8,803.15' from an invoice email. This was achieved through its detailed "Context" (C) component—which specified the email structure and information fields—and its "Polish" (P) component for quality refinement. This success demonstrates CO-STEP's effectiveness in precise data extraction, overcoming both Zero-shot limitations (where background information is lacking) and Few-shot constraints (which depend on similar examples). While CO-STAR handles basic scenarios, its "Objective" (O) and "Response" (R) components lack the detailed control that CO-STEP's "Execution" (E) component provides through specific formatting instructions.

IV. CONCLUSIONS

In this study, we introduced CO-STEP, a novel prompt engineering framework designed to enhance the performance of LLMs like OpenAI 4o-mini. CO-STEP integrates clear goal setting, detailed context, task breakdown, chain-of-thought prompting, and iterative refinement to optimize LLM interactions. Experimental results demonstrate that CO-STEP achieves superior performance, with an Accuracy of 0.98 and an F1-Score of 0.97, making it highly effective for tasks like data extraction and mathematical reasoning.

Despite its promising results, CO-STEP has important limitations. While it excels at structured tasks like data extraction and mathematical reasoning, it struggles with ambiguous prompts and creative tasks requiring open-ended responses. The framework's reliance on detailed context and iterative refinement also increases computational costs and processing time, creating challenges for real-time applications. Future research should explore CO-STEP's adaptability to creative tasks like text generation and storytelling. Testing CO-STEP with other LLMs beyond OpenAI 4o-mini, including Meta's LLaMA [10], Google's Gemini, and xAI's Grok, will be crucial to validate its broader applicability. Research should also focus on developing optimization strategies to reduce CO-STEP's computational overhead, making it more suitable for real-time applications.

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